

- 20 Discuss the way in which scientific conclusions about controversial issues, such as what actions should be taken to reduce global warming or the degree to which humans are affecting global warming, can sometimes depend on who is reaching the conclusions.
- 21 Describe how evolution (a change in the allele frequency) can come about through gene mutation and natural selection.
- 22 Explain how reproductive isolation can lead to speciation.
- 23 Describe the role of the scientific community in validating new evidence (including molecular biology, e.g. DNA, proteomics) supporting the accepted scientific theory of evolution (scientific journals, the peer review process, scientific conferences).

3.4 Topic 6: Infection, immunity and forensics

Students will be assessed on their ability to:

- 1 Demonstrate knowledge and understanding of the *How Science Works* areas listed in the table on page 12 of this specification.
- 2 Explain the nature of the genetic code (triplet code, non-overlapping and degenerate).
- 3 Explain the process of protein synthesis (transcription, translation messenger RNA, transfer RNA, ribosomes and the role of start and stop codons) and explain the roles of the template (antisense) DNA strand in transcription, codons on messenger RNA, anticodons on transfer RNA.
- 4 Explain how one gene can give rise to more than one protein through post-transcriptional changes to messenger RNA.
- 5 Describe how DNA profiling is used for identification and determining genetic relationships between organisms (plants and animals).
- 6 **Describe how DNA can be amplified using the polymerase chain reaction (PCR).**
- 7 **Describe how gel electrophoresis can be used to separate DNA fragments of different length.**
- 8 Distinguish between the structure of bacteria and viruses.
- 9 Describe the role of micro-organisms in the decomposition of organic matter and the recycling of carbon.
- 10 Describe the major routes pathogens may take when entering the body and explain the role of barriers in protecting the body from infection, including the roles of skin, stomach acid, gut and skin flora.
- 11 Explain how bacterial and viral infectious diseases have a sequence of symptoms that may result in death, including the diseases caused by *Mycobacterium tuberculosis* (TB) and Human Immunodeficiency Virus (HIV).

- 13 Explain the roles of antigens and antibodies in the body's immune response including the involvement of plasma cells, macrophages and antigen-presenting cells.
- 14 Distinguish between the roles of B cells (including B memory and B effector cells) and T cells (T helper, T killer and T memory cells) in the body's immune response.
- 15 Explain how individuals may develop immunity (natural, artificial, active, passive).
- 16 Discuss how the theory of an 'evolutionary race' between pathogens and their hosts is supported by the evasion mechanisms as shown by Human Immunodeficiency Virus (HIV) and *Mycobacterium tuberculosis* (TB).
- 17 Distinguish between bacteriostatic and bactericidal antibiotics.
- 18 **Describe how to investigate the effect of different antibiotics on bacteria.**
- 19 Describe how an understanding of the contributory causes of hospital acquired infections have led to codes of practice relating to antibiotic prescription and hospital practice relating to infection prevention and control.
- 20 Describe how to determine the time of death of a mammal by examining the extent of decomposition, stage of succession, forensic entomology, body temperature and degree of muscle contraction.